REMEDIAL WORKS OF AUTOMATED SPRINKLER SYSTEM

EMERGENCY CALL OUT TO INVESTIGATE REPORTED ISSUE WITH SPRINKLER SYSTEM

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Template Document Control Table

Ref:	Rev	Issue Date	Comment	Next Review Date
RMR SMS 017	A	July 2024	New complete SSoW to use as a template for all Service RAMS	July 2025
	B	November 2024	Updated COSHH and added working around batteries controls, PPE & emergency procedures	November 2025

Site Specific Issue Record

Doc Ref	Author	Date Issued	Version	Comments
RMR SMS 017				

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1. Project Detail

Main contractor	Alpine Fire Engineers Ltd	0161 791 4500
Site Name	Amazon BHX5 Rugby	
Attending Service Engineer	Liam Goodman Brett Sutton	
Site Contact details	Paul Robinson	
Engineer Manager Contact	Aaron Coombes	
HSEQ Manager	Tracey A Golden	07741 653 961

2. Scope of Work

This method statement and all associated risk assessments are for sprinkler service works and any remedial works.

Any requirements to deviate from this safe system of work will be re-assessed and further controls implemented.

3. Responsibilities

Alpine Service Manager	Ensuring this SSoW has been reviewed & implemented for all works within its scope with all the necessary provisions & is briefed to & understood by the Service Team. Checking the team are competent to carry out the tasks within this SSoW.
Senior Service Engineer	Ensuring that the Service Engineer is supported & understands this SSoW & that it remains suitable to the task within the sequence of work. Ensuring all the mandatory pre-start checks are being completed & that tools & equipment provided is suitable & sufficient for the work. Assisting in developing safe systems of work with the Engineers in their management & liaising with all parties to report shortfalls for continual improvement. Providing the SHE communication & consultation to the Service Engineer team being the first point of contact for any issues.
Service Engineer	Ensuring that a point of work risk assessment is produced on arrival to check this SSoW is suitable & sufficient to control any risk. To adhere to this SSoW & stop immediately following any deviation or additional risk or if feeling unsafe at any time & reporting to the Client & Alpine Senior Service Engineer. Reporting any incidents, hazard spots or near misses immediately.
Service Department Team	Supporting the Client & Service Engineers in the collation & issue of all necessary RAMS & associated documents liaising with SHE & the Service Managers.
The Client	The Clients designated person will carry out a site induction to ensure that all Engineers have been briefed regarding site specific safety, security, and emergency procedures prior to commencing work on site. The Client will ensure all welfare arrangements are made available for the visiting Engineer.

4. General Health & Safety

4.1 Covid-19

This method statement will be issued alongside the current government guidance relating to Covid 19 all other Clients C-19 policies and controls provided at site induction which will be always adhered to.

4.2 Competence

All Engineers will hold the necessary competencies for the task and any work equipment which will be issued to the Client pre-commencement by the Service Department.

4.3 Arrival & Induction

Alpine Engineers will arrive in a company branded vehicle in a safe manner and will adhere to all route and speed limit rules in accordance with the site security and/or signage.

They will sign in at reception and receive a site-specific induction on arrival and prior to attending the workface. They will adhere to all site rules issued and obtain any necessary permit to work from the Client as requested paying particular attention to pedestrian and mobile plant/vehicular routes.

4.4 Tools & Equipment

Type	Necessary checks	To be completed by
General hand tools	Daily visual	Service Engineer/User
Company vehicle	Daily visual weekly recorded	Service Engineer/Driver
MEWP	Daily visual recorded 6 monthly LOLER	Service Engineer/User Third party

All tools and equipment used will be suitable for task, used only as intended, defect free and sufficient. Any defective tools or equipment will be immediately removed and replaced.

4.5 Access / Egress to site

The Engineer will access the work area always showing situational awareness regarding the routes, possible slip, trip or fall hazards, moving plant, third party activities and other equipment within the area.

4.6 Asbestos

Although it is understood that ACM's are not prevalent on this project, the Engineer will be informed of any possibility of asbestos at induction stage as per the Client asbestos register (where applicable).

All Engineers will hold a UKATA Asbestos Awareness certificate, complete a dynamic risk assessment (POWRA) prior to commencement and will receive a copy of the Alpine Asbestos Discovery Plan as standard.

Wherever the Engineer suspects that any materials are asbestos, or may contain asbestos, they will stop work immediately and report this to the Client Site Manager and the Alpine Service Manager/HSEQ department.

Work will not re-commence until the possible asbestos or ACM has been investigated and has been advised is safe or has been removed by a licenced contractor.

Disturbing or removing possible ACM's is strictly prohibited for any Alpine Engineer.

4.7 Working at Height (Tanks with fixed lifeline) - where applicable

If it is necessary to climb the ladder to access the inspection platform at the top of a tank, this must be done so in the presence of someone who is prepared to raise the alarm and contact emergency services in the event of an emergency. If this is not possible, only a visual check of the contents gauge can be completed.

(N: B In all circumstances, no work at height on tanks will be completed in high winds and adverse weather conditions and a return visit will be scheduled)

- The condition of the ladders to access the tank will be visually inspected by the Service Engineer on arrival and access to the top of the tank will only be carried out if they are happy the access ladders are secured to the side and defect free.
- Wherever tank valve inspections are required/safe to complete, the tank ladders will be carefully removed from the pump house using correct manual handling techniques and secured in place using the hooks at the bottom of the tank ladder cage.
- The Service Engineer will use their personal harness and a suitable fall "arrest" lanyard in all instances where tank inspections are required, and a lifeline is in place.
- Only once the ladder is secured and the Engineer has donned their personal PPE will they access the ladder.
- The Engineer will clip onto the lifeline at the lowest point and not remove their lanyard contact from this all the way to the top of the tank ladder.
- Once at the top, the Engineer will balance and secure their footing and only when happy that the platform is secure, and they feel balanced will they unclip their lanyard to clip onto a suitable anchor point to complete the ball valve check.
- Once completed the valve hatch will be closed and secured and the Engineer will move through the platform gate and immediately clip back onto the lifeline.
- Once clipped on, they can then make their decent down the ladder always maintaining 3 points of contact.
- Once at the bottom of the lifeline, they will then unclip themselves and carefully descend to ground level.
- The bottom ladder access will always be removed following the test and inspection and placed back into the designated area (usually the pump house) in a safe place to prevent trips and unauthorised use of the ladders.

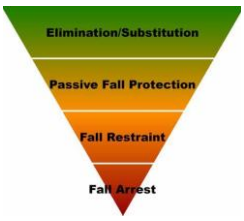
4.8 Working at Height (Tanks with no fixed lifeline) - where applicable

- The same controls above (4.6) will be followed without the use of a fall arrest system when climbing the ladder. Once at the inspection platform, Engineer to clip onto suitable anchor point using fall restraint lanyard & harness.

N.B any anchor points designed into the tank must be suitable and fit for purpose to use.

5. Working at Height Controls

- All work at height will always be suitably planned and controlled considering the hierarchy of control:
- Where implemented, a work at height permit is to be obtained from the Principal Contractor and implemented prior to any work commencing.



STOP!!

- Wherever access equipment is not suitable to reach an area – STOP work immediately and report this to your Senior Engineer / Technical Manager / HSEQ Departmeny who will arrange for more suitable equipment.

DO NOT CONTINUE TO "GET THE JOB DONE" IN ANY INSTANCE WHERE IT IS NOT SAFE TO DO SO!

6. Isolation / impairment of a sprinkler system when carrying out maintenance work or remedial work (where applicable)

The Alpine LOTO and Red Tag procedures will be followed.
The Client is to be informed to enable them to contact their respective insurer to arrange cover amendment.
Everything will be made ready before impairing fire protection equipment i.e. equipment, parts and personnel.

7. Sequence of Work

- Engineer to arrive on site wearing correct PPE as per Section 8 of this document.
- Engineer will inform Control Room prior to work commencing and obtain relevant work permits and inform site contact of the works that are to take place.
- The Client/Customer will be asked to isolate the fire alarm system to prevent any false activations while the works are being carried out,
- The Service Engineer will then carry out Point of Work Risk Assessment and visual pre use checks to ensure to work area is safe to start work.
- Tools, parts, and equipment are to be set up at the work area ensuring all walkways are kept clear and there is no trip hazard around the work area.
- All hand tools are to be visually inspected before use and any tools to be found defective will not be used and replaced. Tools will only be used for their manufactured purpose.
- Small steps will be used for the task for light work only and will only be required for a short duration, all measures will be taken as per section 14.
- Investigate reported issue with sprinkler system.
- Assess issue and rectify if reasonably practicable and task required is a low risk activity with only hand tools required.
- If powered access equipment is required to investigate further then work will need to be rescheduled with appropriate specific Risk assessment and method statements provided.
- Blankets to be placed over all batteries throughout duration of works around pumps.
- Isolate and make safe if necessary, this could include isolating a sprinkler valve or sprinkler pump.
- Close main stop valve on sprinkler system and open main drain valve to drain water from system to prevent any potential damage by water leak if applicable.
- Stop any running pumps and reset to standby mode.
- Attach critical impairment notice to impaired asset and report next steps to rectify issue (if applicable).

- If water storage tank is found to be overflowing, access to investigate float valve can only be completed with a competent person present in attendance, in case of an emergency. If a competent person is not available and/or the issue is not able to be resolved then the tank infill isolation valve will need to be closed and next steps recommended.
- Sprinkler system and associated equipment to be left live and operational where possible, valves to be secured in correct positions with padlocks and straps/chains.
- Blankets can now be removed from the batteries.
- Any impairments to be documented fully and communicated with suitable site representative.
- Remedial request form to be completed to make any required recommendations to rectify any potential issue if required.

5.1 Completion of the Work

- Once completed the Engineer will inform the Client contact that the work is complete including any issues if realised so that the fire alarm can be reinstated.
- The Engineer will then sign off any work permit and sign out of the premises register leaving in adherence to site rules and routes.
- The work area will always be left in a clean and tidy manner as found.

8. Personal Protective Equipment (PPE)

The following PPE is mandatory for all work tasks.



BS EN 388:2016



BS EN ISO 20471



BS EN 345 or 346

For tasks which require working in pump houses & around pumps when running, the below becomes mandatory.



BS EN 166: 2002



EN 352-1:2002

- ✓ Gloves will be a minimum cut resistance level 1 (A)
- ✓ Ear defenders will have a minimum SNR level of 30.
- ✓ Eye protection will be anti-fog and with a minimum low energy impact level - Impact with a small object travelling at 45 meters per second (F)
- ✓ Following a POWRA carried out by the Engineer, establishing there is a possibility of the Engineer striking his/her head on structures above, a bump cap will become a mandatory requirement.



BS EN 812:2012

Wherever POWRA dictates or wherever access to the workface is via boom access, a suitable work restraint / fall restraint lanyard and harness will always be used.

9. Segregation

Wherever an area is not already barriered or protected from other workers whilst testing is being completed, it will be suitably segregated and signed by the Engineer to prevent and warn third parties of the hazards.

10. Emergency Contacts

(The Client's reporting procedure is to be followed in the first instance for all sites. This may include contacting their appointed person initially before the emergency services. Where this is the case, the Engineer will ensure they are familiar with requirements following the site-specific induction process)

All accident / near misses must ALSO be reported to the Alpine HSEQ Department for recording and trend analysis in every instance via the AIMS Incident Reporting Form.

Emergency Services	Fire, Police, Ambulance	999
Environmental Agency	Env Incident	0800 80 70 60

11. First aid treatment

All first aid treatment shall be carried out by the Client's nominated first aider and recorded in the appropriate manner.

All Service Engineers will carry a small first aid box for administering minor first aid to themselves (i.e to clean and add a plaster to a small cut). All SE's will be at least First Aid Awareness trained.

12. Environmental Considerations

In accordance with environmental legislation, extreme care will be taken to ensure the least possible damage is caused to the surrounding environment. Spill hazards or accidental release emergency procedures will be as per the Clients processes stated at site induction and will be reported as per the incident reporting procedures of both the Client and Alpine. The Engineer has overall responsibility for applying care when always working and following on site procedures in the event of an Environmental Emergency.

Pump houses where diesel fuel and engine oil are used and / or stored will be always furnished with a suitable spill kit by the Client.

13. Emergency Arrangements / Safety Measures

Evacuation procedure/route will be agreed with the client before commencement of the contract and details briefed at site induction.
All Alpine Service Engineers will be fully conversant with the pre-issued Emergency Plans which will be issued electronically and accessible on their tablets and used on a project by project basis.

In the event of battery explosion, engineer not to touch any contaminated surfaces or blankets or attempt to remove/dispose of these. Engineer to isolate system where safe to do so and immediately contact Pump Technical Manager & HSEQ Officer to liaise with approved hazardous waste removal company to attend and make area safe as soon as possible. Engineer to inform on site contact of incident and advise pump room is made a no-go area and use signage where necessary until contaminated waste is safely disposed of. Engineer

is required to submit an incident report once are has been made safe. Once area is made safe and hazardous waste correctly disposed of, pump engineer to attend site and survey for remedial works to reinstate damaged pump system as soon as possible.

13.1 Fire Precautions

In the event of a fire raise the alarm (as per instructed at induction stage) and inform the PC supervisor immediately, remove yourself to the muster point as discussed with the Client. Where a site induction has not been provided, it is the responsibility of the Service Engineer to request this information on arrival.

13.2 Accident Reporting Procedure

In the event of an accident notify the site responsible person and the first aider immediately. If electric shock is suspected, isolate supplies before approaching injured person. If trained to do so place the IP in the recovery position.

All injured personnel should report to the allocated person as per your induction details and also report all accidents and near misses to Alpine FE Service Manager or SHEQ Manager as per the internal Accident/Incident Reporting Process.

The Alpine Incident Reporting Procedure will be issued to all Service Engineers within their personal H&S Packs.

In the event of a nuclear alarm or fire alarm, emergency procedures will be followed as stated in the main site induction process.

All accidents/incidents and near misses will be investigated by Alpine FE internal Management.

Copies of all RIDDOR accidents shall be forwarded to the client.

14.Risk Assessments

Please note: In all instances, a point of work risk assessment (POWRA) will be completed by the Service Engineer prior to any work commencing which will provide a site specific (dynamic) assessment of the hazards and allow relevant control measures to be implemented.

Rational

Likelihood	Severity				
	Trivial	Minor Injury	Over 3 Day Injury	Major Injury	Incapacity or Death
Highly Unlikely	1	2	3	4	5
Unlikely	2	4	6	8	10
Possible	3	6	9	12	15
Probable	4	8	12	16	20
Certain	5	10	15	20	25

Working at Height

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Falling from MEWP platform/basket due to miss use.	Engineer	Major Injury Incapacity / Death	4	5	20	Only competent people are to operate any MEWP and must hold an in date IPAF card 3a and 3b. All MEWPS are to be supplied with suitable & sufficient guard rails & toe boards with working guard latches & pins. Where it has been realised that the MEWP cannot gain sufficient height or reach for the task, work must STOP & the issue reported to your Senior or the Alpine HSEQ department. Work will not continue until an alternative safe working method has been agreed by all parties. Where a boom type MEWP is being used, a harness and work restraint lanyard is to be donned and the Operator to be clipped on at all times. Harnesses & lanyards will be subject to recorded pre-use checks (by the user) in all instances. Standing on MEWP guard rails is strictly prohibited in all situations and any breaches will be subject to the Alpine carding policy in every instance.	2	5	10

						No additional platform will be added to any MEWP basket to raise the height at any time (i.e. pallets, barrels, ladders etc). Any breaches will be subject to the Alpine (and/or PC) carding policy in every instance. MEWPS will be used in a safe, quiet and slow manner at all times. When using MEWPS mobile phones and ear phones (including wireless) will not be used. Workers are to work within the confines of the platform of the MEWP at all times and will not over stretch, or hang over the edge at anytime. Operatives will wear good sure grip safety boots at all times and will ensure laces are fastened securely to prevent trips. The MEWP basket will be kept in good order at all times with waste materials and tools stored safely to prevent rolling causing trip hazards. MEWPS will always be used as per the machine Operator guidance manual.			
Falling from height due to overturning / tipping of the MEWP.	Engineer	Major Injury Incapacity / Death	4	5	20	A POWRA is to be completed prior to any MEWP use and ground conditions checked. The MEWP will not be used where the POWRA has found that ground conditions are not suitable. In this instance work must STOP & the issue reported to your PE or the Alpine HSEQ department. Work will not continue until an alternative safe working method has been agreed by all parties. Localised ground features (excavations, manholes, uncompacted backfill etc) are to be secured with a suitable, temporary ground cover (suited to max weight of MEWP & load) or avoided completely in the first instance. Load on the MEWP will be safely distributed throughout the basket and not all at one end.	2	5	10

						Where ever possible or needed outriggers are to be fully extended and chocked before raising the platform. MEWPS will always be used in a suitably segregated area at all times to prevent other persons and mobile plant entering the area and bumping the machine. Spreader plates may be required - always check the manufacturers guidance manual prior to use). The MEWP basket will not be overloaded with persons and/or tools & equipment and the SWL will be adhered to at all times.			
Entrapment & crush injuries caused by MEWP ascending too far.	Engineer	Major Injury Incapacity / Death	3	5	15	MEWPS are to be subject to a pre-use check by the Operator including audible & visual warning systems and are not to be used if defective. All MEWPS will be supplied with a valid, in date LOLER certificate and User Guide and will not be accepted/used without these in place and checked. Operators sleeves will be tied close to the skin & not hang loose to prevent them catching controls. Where ever possible, MEWPS to have shrouded controls and anti-entrapment devices including, but not limited to, sensor bars & distance sensors. All MEWPS are to be fitted with anti entrapment audible warning systems. The position of the MEWP must always be so the auxiliary rescue controls at ground level are easily accessible in the event of an emergency. A POWRA is to be completed prior to use taking into consideration overhead structures and obstructions which are to be pointed out to the team before ascent. Before raising and lowering the platform, the Operator must check all 4 sides of the MEWP (up and down) to check for obstructions/protrusions which may catch.	1	5	5

						A robust rescue plan must be in place before use.			
Collision with persons, equipment, materials and structures on the ground.	Engineer Any third party	Property Damage Major Injury Incapacity / Death	3	5	15	All other personnel / contractors / third parties are to be segregated from the area by clearly signed and barriered areas and pedestrian walkways. No unauthorised personnel are to be allowed within the work area. The Operator is to use due diligence at all times when manoeuvring the MEWP and check all sides before starting to ensure the route is clear. Where ever possible a banksman on spotter at ground level is to be used and will stand away from the MEWP to be visible with high visibility clothing on.	1	5	5
Collision or fall from height due to incompetence / in experienced Operators.	Engineer Any third party	Property Damage Major Injury Incapacity / Death	2	5	10	Only competent persons are to Operate any type of MEWP. Do not allow others to use any MEWP supplied by/for/on behalf of Alpine before first checking with the HSEQ Manager AND Project Engineer. Do not use another Contractors MEWP without written consent from them, plus completing your own pre-use, LOLER & user guide check. Inexperienced Operators must be "buddied" with more experienced persons until deemed confident and competent by the Foreman. Familiarisation training will be given to any inexperienced persons and to the team when a different machine is delivered. The Operators must ask for this from a competent person if they are not familiar with a new machine.	1	5	5

Working around batteries (lead acid & NiCad)

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Explosion / short-circuit due to conductive	Engineer, Any third party		4	5	20	Reduce time around batteries wherever possible. Area to be well ventilated, with doors remaining open throughout works.	2	5	10

materials coming into contact with battery terminals causing failure and fire						No sources of ignition (e.g. sparks, flame) to be present when working near batteries. All tools to be insulated. Do not place tools or cables on batteries or in an area where they can fall onto the terminals. No unauthorised persons to be within the area when diesel pump is to be started, during the run time, and when the pump has stopped. All engineers to be suitably trained for working around batteries. Visual checks to be completed prior to work commencing to ensure no corrosion, leaks, swelling, cracks and to ensure batteries are dry and free from any liquid. Where engineer is trained to do so, battery levels to be checked and topped up where necessary before work commencing. Engineers to be suitably briefed on the Emergency Spill Response. If batteries are older than working life (Lead Acid 5 years, Nicad 10 years) engineer to stop work and submit replacement quote request. Correct PPE to be worn whenever working around batteries including butyl gloves and visors. Battery blanket to be placed over batteries when diesel pump is to be started, during the run time, and when the pump has stopped, ensuring all the sides of the batteries are covered and there are no kinks or creases in the blanket. Adequate first aid facilities and equipment to be situated locally to the work area including eye wash.			
Spills and leaks from batteries	Engineer, Any third party		3	4	12	All engineers to be suitably trained for working around batteries. Visual checks to be completed prior to work commencing to ensure no corrosion, leaks, swelling, cracks and to ensure batteries are dry and free from any liquid.	2	4	8

						If engineer notes any damage to the batteries, batteries to be isolated and impaired to prevent any third parties or other engineers from exiting the batteries, Engineers to be suitably briefed on the Emergency Spill Response. Correct PPE to be worn whenever working around batteries including butyl gloves and visors. Battery blanket to be placed over batteries when diesel pump is to be started, during the run time, and when the pump has stopped, ensuring all the sides of the batteries are covered and there are no kinks or creases in the blanket.			

Using Pipe Threading Machines

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Manual handling injuries such as burns, skin abrasions, cuts, repetitive strain, muscle pulls and strains	Engineer	Minor Injury Major Injury	3	4	12	Correct PPE to be worn at all times when using the threading machine. Persons using the machine to stretch and warm up prior to use. A comfortable and correct working position to be decided and maintained throughout the threading work. Suitable fire extinguishers and a first aid kit to be present at all times whilst machine is in use. Work area is to be suitable with minimal interference whilst using the machine.	2	4	8
HAVS (hand, arm, vibration) white finger, carpel tunnel, repetitive strain	Engineer	Minor Injury Major Injury	3	4	12	Where ever possible the use of the threading machine will be kept to a minimum and no more than the exposure limit advised on the machine. Threading only to be carried out in short duration bursts with Operators rotating use in line with HAVS guidance's which will be recorded on the AFE HAVS Register.	2	4	8

						Threading machine to be fit for purpose, well maintained and defect free at all times and fitted with sufficient viubration suppression devise. Hands are to be kept dry and warm at all times when using the threading machine. Operators are to stop work immediately if they are suffering with tingles, numbness or pins and needles in the fingers, hands or arms.			
Moving components / machine parts causing entrapment of hair & clothing	Engineer	Minor Injury Major Injury Incapacity / Death	2	5	10	Operator of threading machine to be trained and competent to use. No unauthorised personnel are to use machine. All machines to be issued with safety guards and cut offs. Any guards issued within the interests of health and safety not to be tampered with or removed under any circumstance. Good fitting work gloves, safety goggles and ear defenders to be worn as a minimum when using the threading machine. Loose handing jewellery (necklaces), rings, bracelets and watches to be removed prior to use. No loose or baggy clothing (especially sleeves) to be worn and long hair to be tied back and ties or lanyards are not permitted when using threading machine. when using the machine. Threading machine to be subject to pre-use checks with all defects reported immediately and the machine removed from use until fixed or replaced.	1	5	5

Noise

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Noise from running pump	Engineer Any third party	Minor Injury Major Injury	5	4	20	No person to enter the immediate area when a pump is running or is to be switched on unless necessary. If necessary, hearing protection of a minimum SNR30 are to be always used.	2	4	8

Noise when using threading machines	Engineer Any third party	Minor Injury Major Injury	2	4	8	Noise zones to be established in accordance with current legislation and ear defenders supplied in the 1st action level zone (80db) on request. Where the 2nd action level (85db) is reached, ear defenders will become a mandatory requirement for all personnel within the work area. Operatives use of the threading machine will be kept to a minimum at all times.	1	4	4

Access / Egress / Slips, Trips, Falls

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Uneven ground Loose tools. Rubbish. Moving plant & machinery. Wet floor conditions.	Engineer Any third party	Minor Injury Major Injury	3	4	12	Area to be always left in clean and tidy state but all premises' users. Engineer to use point of work risk assessment to take 20 and check area. Lighting to be maintained and suitable within all areas by Client. Engineer to have received short duration training for situational awareness. Safety boots with tied laces and sure grip soles to be always worn by the Engineer. Engineer to receive a site-specific induction on arrival to be informed of all vehicular and mobile equipment routes and pedestrian walkways which will be always followed. Engineer to continually wear high visibility clothing whilst on the premises which will be donned before accessing and not removed until exiting the private premises. Any water leakage will be soaked away immediately (before work commences if found on arrival) and the area left clean and dry.	1	4	4

						Any issues found on arrival at the workface are to be reported immediately to the Client and controlled to prevent risk before testing commences.			
Foam spill causing slip and trip hazard	Engineer Any third party	Minor Injury Major Injury	3	4	12	Any foam spill is to be identified immediately and contained using the spill kits that is out and ready in the work area.	2	4	8
Wires and leaks when using machinery	Engineer Any third party	Minor Injury Major Injury	3	4	12	Where possible cables are to be run above head height or if not possible, kept tidy to prevent trips. Cables are to be uncoiled completely when machine is in use. Where ever possible a plastic drip tray will be used in lieu of plastic sheets and ply to reduce the risk of trips, slips or falls.	1	5	5

Working in live premises

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Other workers Moving equipment	Engineer Any third party	Minor Injury Major Injury	3	4	12	Where others are working in or around the Alpine workface, segregation and signage will be used. Alpine Engineer will be made aware of all others working in the area and zoned sections of the building and will adhere to all rules issued at induction stage. The Alpine work area will be kept clean and tidy whilst testing progresses and will be left as found on completion. Where the Engineer feels their safety has been put at risk or their work may put others at risk, they will immediately stop and report the hazard to allow for controls to be implemented. No unauthorised persons are to work closely to where the Engineer is testing.	2	4	8

Pressurised systems

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)

Water escape under pressure. Projectile fittings. Water leakage.	Engineer Any third party	Minor Injury Major Injury	2	4	8	All Engineers to wear suitable safety glasses when testing. No unauthorised persons to be within the area when testing commences. Any water escape is to be dealt with to ensure the area does not produce a slip hazard.	1	4	4
Foam concentrate escaping under pressure	Engineer Any third party	Minor Injury Major Injury	2	4	8	All Engineers to wear suitable safety glasses when working around or on foam system or there components. Foam system drains to be checked and not running before work is to start. All gauges on the system are to be checked and showing no pressure before work will commence.	1	4	4

Manual Handling

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Seized and stiff valves. Carrying tool kits.	Engineer	Minor Injury Over 3-day absence	2	4	8	All Engineers to have received manual handling training. Tools are to be defect free and suitable and sufficient for the task. Tools are to be kept to a minimum to prevent the necessity for carrying heavy toolboxes. Sure grip gloves to be work when exercising any valves or pumps handles. Where an element has seized or overexertion would be required to movie/open it, the Engineer will stop and report this as a defect and not over try to manually loosen the item.	1	4	4

Safe use of Steps & Ladders

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Working from height with potential to fall	Ladder / Step user	Over 3 day injury; Major Injury;	3	5	15	Only persons deemed competent are to use ladders of steps. Ladders or steps only to be used for low risk, short duration work.	1	5	5

		Incapacity / Death				Only ladder and steps which are BS EN 131 standard approved will be used. Point of Work Risk Assessment is to be completed prior to use of any staps or ladders to assess the work environment and suitability of the equipment. Ladders or steps are only to be used for the purpose they are intended. Visual pre-use checks will be made in every instance and any defects will be noted/recorded and the ladder/steps removed from use and tagged until fixed or replaced. Any ladder/steps that have been painted will be immediately removed from use and replaced. Ground conditions and immediate environment to be checked for suitability before use. Wherever required by the Client, a ladder permit and tag system will be used. Ladders / steps will be store horizontally at all times when not in use. Water tank ladders will be removed and securely stored when not in use.			

Safe use of Portable Hand Tools

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Manual handling injuries such as burns, skin abrasions, cuts, repetitive strain, hand and arm vibration.	Equipment / tool user	Over 3-day injury; Major Injury.	2	4	8	All Operators/users are to be trained and competent. No unauthorised personnel are to use any tools and equipment. All electrical equipment to be used with safety guards and cut offs. Good fitting work gloves to the correct cut/shred rate are to be always used. No loose or baggy clothing (especially sleeves). Long hair is to be tied back.	1	4	4

						Suitable fire extinguishers are to be within easy reach. A comfortable and correct work position is to be used. Suitable safety goggles are to be always used. All electrical equipment is to have a suitable scheduled PA Test. All handheld power tools are to be suitable for the task and defect free and are subject to daily, visual pre-use checks and monthly thorough, recorded checks by the user. Safety guards are to remain in place and must not be tampered with. Work areas are to be kept dry and tidy with liquids stored away from power cables and generators. Electricity will run at no more than 110v on site and all power sources will be set with breakers as standard.			

Working Where There is Risk of Falling Materials

Hazard	Persons at Risk	Possible outcome	(L)	(S)	R	Controls	(L)	(S)	(RR)
Tools, equipment and materials falling from height (work platforms) striking persons below work area.	Contractors, third party visitors, public (all at ground level)	Minor injury, major injury, single fatality	3	4	12	No unauthorised persons allowed on site at any time. All Contractors to received a site specific induction prior to attending site. All persons working within the segregated drop zone around any Alpine work **MUST** wear a suitable hard hat at all times. **A safety zone is to be put in place to prevent unauthorised personnel from accessing the work area where persons are working at height. (Please refer to the deflection diagram below when assessing the radius required of any drop zone).** Work safety drop zone will be large enough, taking into account bounce from force, clearly signed and guarded at all times using cones and warning tape as a minimum to ensure manoeuvrability of access equipment remains practical.	2	4	8

						Where suitable segregation is not possible (i.e within some rack systems, or in conjested areas), suitable tool tethers become a **MANDATORY** requirement. Hard hat chin straps will be **MANDATORY** in all areas where work at height is being carried out. All access equipment to be fitted with secure toes boards. Sure grip gloves to be worn when carrying out work tasks. Where possible tool belts to be worn. When working more than 2mtrs from a leading edge, hard hat chin straps are to be used at all times. Access platforms (baskets) to be kept clear from debris at all times with good order/housekeeping maintained. Access equipment (MEWPS) to be supplied with manufacturer installed pipe cradles. Where this is not possible, Unistrut will be used to construct a cradle until one can be fitted or supplied by the plant supplier. Falling object near misses are to be reported immediately to the HSE Manager. Any future trends may result in the use of tool tethers/lanyards being used as a mandatory requirement. Access under working platforms or where personnel are carrying out work at height is to remain strictly prohibited. Edges will be fitted with edge protection or fall prevention i.e. guard rails and toe boards, netting etc.			

15.COSHH

1. AFE CA 039 Lead Acid Multicell Flooded Battery
2. AFE CA 038 Electrolyte ALCAD

COSHH ASSESSMENT – Electrolyte ALCAD

Material/Product:	Electrolyte ALCAD		
Manufacturer/Supplier:	Alcad LTD, Norra Strandgatan 35, Box 504, S-572 28 Oskarshamn, Sweden		
Composition:	Colourless, odorless fluid		
Area/Location:	Used in industrial Ni-Cd-batteries on fire suppression pumpsets	Assessment No:	AFE CA 038 RevA
Assessed By:	Sadie Henson	Last Review Date:	October 2024
Title:	HSEQ Coordinator	Next Review Date:	October 2025

The Substance – Hazard Identification:
Please denote the properties of the substances used in the task (as per EU D Risk Phrases below)

 Flammable	 Acute Toxicity	 Harmful/Irritant/ Skin Sensitiser	 Corrosive	 Harmful to the Environment
 Carcinogenic / Mutagen / Teratogen	 Compressed Gas	 Explosive	 Oxidising	

The effects of the substance on the body: Take into account any changes in the substances state, during the task	
(EU D) Risk Phrases #	R22, R35, R41
Inhalation	Symptoms include burning, coughing and difficulty breathing. Risk of lung damage in high concentrations.
Ingestion	Ingestion can cause serious corrosive injuries with burning pain, vomiting, stomach pains, possibly severe general effects (shock) and kidney damage. If ingested, severe burns of the mouth and throat, as well as a danger of perforation of the oesophagus and the stomach. Even small amounts may cause serious damage. Risk of permanent injuries
Skin Contact	Strongly corrosive to the skin and mucous membranes. Serious corrosive injuries with slow-healing wounds can occur. Risk of permanent injuries.
Eye Contact	Splashes in the eyes may cause pain and burns. Risk of permanent damage to vision.
First Aid Requirements: Details to be found on the Material Safety Data Sheet	
Exposure via:	**First Aid Treatment:**
Inhalation	In case of inhalation, move the affected person to fresh air and keep at rest in a position comfortable for breathing. If symptoms persist, call a physician
Ingestion	Do NOT induce vomiting. Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Contact your doctor immediately.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical treatment necessary as untreated skin corrosions are slow and bad healing wounds. Call a physician immediately
Eye Contact	In case of eye contact, remove contact lens and rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Contact your doctor immediately. Continue rinsing eyes during transport to hospital
Handling and Storage:	
Handling	Keep container tightly closed. Use protective equipment as specified in PPE section of this

	COSHH Assessment. Avoid contact with the skin and the eyes. Do not breathe vapours or spray mist. Ensure that eyewash stations and safety showers are close to the workstation location.
Storage	Requirements for storage areas and containers : Keep in an area equipped with alkali resistant flooring. Store in original container. Advice on protection against fire and explosion: The product is not flammable. Normal measures for preventive fire protection. Gives off hydrogen by reaction with metals. risk of explosion. Further information on storage conditions : Keep container tightly closed. Store in well-ventilated place. Advice on common storage : Keep away from food, drink and animal feeding stuffs. Do not store together with acids and ammonium salts.

Fire Fighting Measures:

SPECIAL FIRE FIGHTING PROCEDURES: Keep unauthorized persons away. Fire residues and contaminated firefighting water must be disposed of in accordance with the local regulations

SPECIFIC HAZARDS: Contact with certain metals e.g. zinc, lead, tin, aluminium and magnesium and their alloys can generate hydrogen gas (explosive with air). Contaminated surfaces will be extremely slippery.

Suitable Extinguishers: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. The product itself does not burn.

Unsuitable Extinguishing Method: Do not use a heavy water stream

Special Hazards Arising: Contact with certain metals e.g. zinc, lead, tin, aluminium and magnesium and their alloys can generate hydrogen gas (explosive with air). Contaminated surfaces will be extremely slippery

Protective Equipment: Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

HAZARDOUS DECOMPOSITION PRODUCTS: None known

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Exposure Controls/Personal Protection:

Exposure Controls: Ensure that eyewash stations and safety showers are close to the workstation location.

PERSONAL PRECAUTIONS Keep unauthorized persons away. Spill area may be slippery. Avoid contact with the skin and the eyes.
Do not breathe vapours or spray mist. Use protective equipment as specified in PPE section of this COSHH Assessment.

ENVIRONMENTAL PRECAUTIONS Do not flush into surface water or sanitary sewer system. Avoid subsoil penetration. If the product contaminates watercourses, lakes, soil or drains inform respective authorities

SPILL CLEAN UP METHODS Clean up any spills as soon as possible, using an absorbent material to collect it. Collect up spilt material and transfer to a suitable container for re-use or subsequent disposal. Use containers made of plastic.
Do not use metal containers

HYGIENE MEASURES Keep away from food, drink and animal feeding stuffs. Smoking, eating and drinking should be prohibited in the application area. Wash hands before breaks and immediately after handling the product. Remove and wash contaminated clothing before re-use.

Personal Protective Equipment

Respiratory Protection: Use approved respirator if air contamination exceeds acceptable level. If it is suspected that fumes are still present wear an appropriate mask or self-contained breathing apparatus. Recommended filter type: Particle filter : P3



Hand Protection: Use chemical resistant gloves that corresponds to EN 374. Potassium hydroxide-resistant gloves. If signs of wear and tear are noticed then the gloves should be replaced.



Eye Protection: Safety glasses with side-shields conforming to EN166



Protective Clothing: dust impervious protective suit , Alkali-proof protective suit .



Exposure Control Parameters: Potassium hydroxide – Short-term exposure limit 2 ppm / mg/m³
Lithium hydroxide– Short-term exposure limit 1 ppm / mg/m³

Environmental Exposure Controls

Accidental spills: Clean up any spills as soon as possible, using an absorbent material to collect it. Collect up spilt material and transfer to a suitable container for re-use or subsequent disposal. Use containers made of plastic.
Do not use metal containers.

For large spills: Do not flush into surface water or sanitary sewer system. Avoid subsoil penetration. If the product contaminates watercourses, lakes, soil or drains inform respective authorities.

Disposal Considerations:

Prevent discharges to watercourses, waste water or the ground.
Dispose of any product, residue or packing material according to national and local regulations. Give to a certified disposal contractor.
Empty the packaging thoroughly. Empty containers can be reused. Packaging that can´t be cleaned are taken care of in the same manner as the product.

Waste code	Description
16 06 06*	separately collected electrolyte from batteries and accumulators

Please note – an asterisk (*) next to a code denotes that it is HAZARDOUS WASTE.

Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated firefighting water must be disposed of in accordance with the local regulations.

COSHH ASSESSMENT – Lead Acid in Batteries

Material/Product:	Lead Acid Batteries		
Manufacturer/Supplier:	Multicell International Ltd, Box Road, Broughton Astley, Leicester, LE9 6TJ		
Composition:	Off white cloudy liquid with solid object.		
Area/Location:	Within Batteries for Firefighting Pumpsets	Assessment No:	AFE CA 039 RevA
Assessed By:	Ellie Monaghan	Last Review Date:	Oct 2024
Title:	HSEQ Officer	Next Review Date:	Oct 2025

The Substance – Hazard Identification:
Please denote the properties of the substances used in the task (as per EU D Risk Phrases below)

 Flammable	 Acute Toxicity	 ✓ Harmful/Irritant/ Skin Sensitiser	 ✓ Corrosive	 Harmful to the Environment
 ✓ Carcinogenic / Mutagen / Teratogen	 Compressed Gas	 Explosive	 Oxidising	

The effects of the substance on the body:
Take into account any changes in the substances state, during the task

(EU D) Risk Phrases	R20, R23, R35, R41, R46, R49, R48/25; R48/24; R48/23
Inhalation	Not a likely route of exposure. Corrosive severe irritation and burns. If a battery ruptures may be harmful or fatal if inhaled in a confined area. May cause severe irritation and burns on the nose, throat and respiratory tract.
Ingestion	Not a likely route of exposure. Causes serious burns of the mouth or perforation on the oesophagus or stomach. May be fatal if swallowed.
Skin Contact	Not a likely route of exposure. Direct contact with internal components of a battery can be severely irritation to the skin and may result in redness, swelling, burns and severe skin damage. Skin contact may aggravate an existing dermatitis condition.
Eye Contact	Not a likely route of exposure. If a battery ruptures, direct contact with the liquid or exposure to vapours or mists may cause tearing, redness, swelling and corneal damage and irreversible eye damage. Splashes in the eyes will cause severe burns.

First Aid Requirements:
Details to be found on the Material Safety Data Sheet

Exposure via:	First Aid Treatment:
Inhalation	If a battery ruptures, move to fresh air in case of accidental inhalation of mist. If breathing has stopped, perform artificial respiration. If breathing is difficult, give oxygen. GET MEDICAL ATTENTION AS SOON AS POSSIBLE. Remove victim to fresh air and keep at rest in a position comfortable for breathing
Ingestion	If solutions of a battery chemicals have been swallowed and the person is conscious, give one glass of water. Vomiting may occur spontaneously, but Do NOT induce vomiting. Never give anything by mouth to an unconscious person. GET MEDICAL ATTENTION IMMEDIATELY. Rinse mouth. Do NOT induce vomiting.
Skin Contact	If a battery ruptures, do not rub or scratch exposed skin. If liquid get on the skin, immediately flush the contaminated skin with water for at least 15 minutes. If liquid penetrate through the clothing, immediately remove the clothing and shoes under a safety shower and continue to wash the skin for at least 15 minutes. GET MEDICAL ATTENTION IMMEDIATELY. Remove/Take

	off immediately all contaminated clothing. Rinse skin with water/shower.
Eye Contact	If a battery ruptures, do not rub or scratch exposed eye. Immediately flush eyes with running water for at least 15 minutes, keeping eyelids open. Cold water may be used. GET MEDICAL ATTENTION IMMEDIATELY. Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
Handling and Storage:	
Handling	Protect from physical damage.
Storage	Avoid contact with eyes. Store in a cool, dry, ventilated area away from sources of heat, moisture, incompatibilities, and direct sunlight. Have emergency equipment (for fires, spills, leaks, etc.) readily available.
Fire Fighting Measures:	
UNUSUAL FIRE & EXPLOSION HAZARDS Not flammable. Battery may rupture due to pressure buildup when exposed to excessive heat and may be result in the release of corrosive materials. SPECIFIC HAZARDS Lead, lead compounds and sulfuric acid fume may be released during a fire involving the product. **Suitable Extinguishers:** Use extinguishing media appropriate for surrounding fire. If a battery ruptures, use dry chemical, soda ash, lime, sand or carbon dioxide. **Protective Equipment:** Wear self contained breathing apparatus (SCBA) and full firefighting protective clothing. HAZARDOUS DECOMPOSITION PRODUCTS If a battery ruptures, avoid contact with organic materials and alkaline materials. Lead, Lead compounds and sulfuric acid fumes may be released during a fire involving the product.	
Exposure Controls/Personal Protection:	
Exposure Controls: Use local exhaust ventilation if necessary to control airborne mist and vapor PERSONAL PRECAUTIONS If a battery ruptures, avoid contact with skin, eyes and clothing. Do not touch spilled material. Use personal protective equipment recommended in Section 8 (Exposure Controls/Personal Protection). ENVIRONMENTAL PRECAUTIONS Notify authorities and appropriate federal, state, and local agencies. Prevent the product from spreading into the environment. Avoid direct discharge into drains. SPILL CLEAN UP METHODS **SMALL SPILLS:** Collect all released material in a plastic lined metal container. If necessary neutralize the residue with a dilute solution of sodium carbonate. Wash affected area. **LARGE SPILLS:** Contain liquid using absorbent material, by digging trenches or by building a dike. Absorb with dry earth, sand or other non-combustible material. Neutralize the residue with a dilute solution of sodium carbonate. Dispose of all contaminated materials in accordance with current local regulations. HYGIENE MEASURES Do not eat, drink or smoke when using this product. Wash contaminated clothing before reuse. Wash hands thoroughly after handling.	
Personal Protective Equipment	
Respiratory Protection: Notify authorities and appropriate federal, state, and local agencies. Prevent the product from spreading into the environment. Avoid direct discharge into drains.	
Hand Protection: Wear chemical resistant gloves. Gloves should be replaced immediately if signs of degradation are	

observed.
Eye Protection: Wear safety glasses with side shields (or goggles).
Protective Clothing: Use good work and personal hygiene practices to avoid exposure. Consider the provision in the work area of a safety shower and eyewash. Always wash thoroughly after handling chemicals.
Exposure Limits: 0.05 mg/m3 (Lead), 1 mg/m3 (Sulfuric acid), 0.5 mg/m3 (Antimony)
Environmental Exposure Controls
Do not touch spilled material. Use personal protective equipment. Notify authorities and appropriate federal, state, and local agencies. Prevent the product from spreading into the environment. Avoid direct discharge into drains. **Accidental spill:** **SMALL SPILLS:** Collect all released material in a plastic lined metal container. If necessary neutralize the residue with a dilute solution of sodium carbonate. Wash affected area. **LARGE SPILLS:** Contain liquid using absorbent material, by digging trenches or by building a dike. Absorb with dry earth, sand or other non-combustible material. Neutralize the residue with a dilute solution of sodium carbonate. Dispose of all contaminated materials in accordance with current local regulations.
Disposal Considerations
Dispose of in accordance with local, state, and federal regulations. Hazardous wastes must be transported by a licensed hazardous waste transporter and disposed of or treated in a properly licensed hazardous waste treatment, storage, disposal or recycling facility. Consult local, state, and federal regulations for specific requirements. **DISPOSING OF CONTAMINATED CONTAINER OR PACKAGE:** Since emptied containers retain product residue, follow label warnings even after container is emptied.